

Table 2: Rules of Natural Deduction	
Introduction Rules	Elimination Rules
$\frac{\phi \quad \psi}{\phi \wedge \psi} \wedge I$	$\frac{\phi \wedge \psi}{\phi} \wedge E \quad \frac{\phi \wedge \psi}{\psi} \wedge E$
$\frac{\phi}{\neg \neg \phi} \neg \neg I$	$\frac{\neg \neg \phi}{\phi} \neg \neg E$
$A^* \quad \frac{\boxed{\begin{array}{c} \phi \\ \vdots \\ \psi \end{array}}}{\phi \rightarrow \psi} \rightarrow I$	$\frac{\phi \quad \phi \rightarrow \psi}{\psi} \rightarrow E$ $\frac{\neg \psi \quad \phi \rightarrow \psi}{\neg \phi} MT$
$\frac{\phi}{\phi \vee \psi} \vee I \quad \frac{\psi}{\psi \vee \phi} \vee I$	$\phi \vee \psi \quad \boxed{\begin{array}{c} \phi \\ \vdots \\ \chi \end{array}} \quad \boxed{\begin{array}{c} \psi \\ \vdots \\ \chi \end{array}} \quad A^* \quad \frac{}{\chi} \vee E$
$A^* \quad \frac{\boxed{\begin{array}{c} \phi \\ \vdots \\ \perp \end{array}}}{\neg \phi} \neg I$	$\frac{\phi \quad \neg \phi}{\perp} \neg E \text{ also } \perp I$
$\frac{}{\top} \top I$	$\frac{\perp}{\phi} \perp E$
$\frac{\phi \rightarrow \psi \quad \psi \rightarrow \phi}{\phi \leftrightarrow \psi} \leftrightarrow I$	$\frac{\phi \quad \phi \leftrightarrow \psi}{\psi} \leftrightarrow E$ $\frac{\psi \quad \phi \leftrightarrow \psi}{\phi} \leftrightarrow E$

Other Rules	
$ \begin{array}{c} A^* \\ \hline \boxed{\begin{array}{c} \neg\phi \\ \vdots \\ \bot \end{array}} \\ \hline \phi \quad \text{PBC} \end{array} $	$ \frac{}{\phi \vee \neg\phi} \text{LEM} $

$$\begin{array}{c}
 A^* \\
 \hline
 \boxed{\begin{array}{c} \neg\psi \\ \vdots \\ \neg\phi \end{array}} \\
 \hline
 \phi \rightarrow \psi \quad \text{CP}
 \end{array}$$

$$\frac{\phi \vee \psi \quad \neg\phi}{\psi} \text{DS} \qquad \frac{\phi \vee \psi \quad \neg\psi}{\phi} \text{DS}$$

$$\frac{}{\phi \vee \neg\phi} \text{EM}$$

$$\frac{\psi \quad \neg\psi}{\phi} \text{EXP}$$

Assumptions are part of the rules. You can't just assume whenever you want.

A^* marked rules are the only rules that allow you to assume.